

Jianxiang(Joseph) Liao

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WORK EXPERIENCE

Network Engineer / Automation System Developer

Los Angeles, CA, US

China Telecom Americas, Network and Operation Center

Aug 2023 – Now

- Provide DevOps, Automation and Datacenter Operations for Physical Line (OTN, SDH...), Ethernet (VLANs, Network Interface...), IP (BGP, IGP, Firewall, Traffic Analysis...), MPLS VPN on OSI Layer 1/2/3 technologies.
- Compile scripts & design tasks for layer 1-3 devices. Configure IP services and perform health checks, troubleshoot.
- Designed and developed an automated admin system to centrally monitor AS36678 network devices (10 cages and 100+ network devices around U.S.). Features included real-time topology visualization, device & circuit health alarms, event logging, batch command and configuration execution, and RPKI-based prefix validation. Added management modules for automated email notifications, secure file storage, and CDKey/license management.
- Built a collaboration and asset-management system using Django Admin and PostgreSQL. Implemented RESTful CRUD APIs (MVC pattern) for device and circuit records, supporting advanced search, operational logging, rack/cage topology mapping, and UPS SNMP trap alarms. Enabled daily NOC operations such as milestone tracking, project progress monitoring, and task assignment.
- Developed a scheduled automation framework (Linux crontab + Python) to parse vendor circuit hotcut emails. Integrated with the internal database for classification, filtering, forwarding, and alerting. Automated generation of structured reports, significantly reducing manual effort and ensuring timely notifications across teams.

Backend & Data Engineer

Pittsburgh, PA, US

Harvard Management Company

Feb, 2022 – May, 2023

- Extracted and analyzed data from 1,200+ hedge fund reports using Kaggle, SEC.gov, and IRS.gov. Built ETL pipelines, designed data models, applied prediction and relevance ranking, and stored results in MongoDB (collections) and Neo4j (graphs).
- Developed backend APIs for customers to perform fund retrieval, portfolio inquiries, investment suggestions.

Full Stack Software Engineer Intern

Melbourne, VIC, AU

Wisecar, Core Development Team

May – August, 2022

- Developed 20+ components and 50+ APIs for a mobile app in React (Native). Implemented a Keycloak system for OAuth 2.0 (user access control) and JWT token verification.
- Architected the microservices infrastructure and designed RESTful APIs in Golang (Gin) for a web portal, enabling CRUD operations for customer reports and claims in the PostgreSQL database. Deployed it through AWS EC2.

Backend Software Engineer Intern

Guangzhou, China

China Telecom, Internet of Vehicle Team

Jun – Sep, 2020

- Implemented JAVA services in SSM framework to efficiently manage 50 servers in a background data system.

PROJECTS

Service-oriented Backend System for Library

Carnegie Mellon University, PA

- Developed RESTful microservices in BFF pattern and single responsibility principle for a library with AWS RDS.
- Leveraged Kubernetes (EKS), Docker and AWS EC2 to deploy and manage microservices. Facilitated system scalability and robustness by load balancers, security groups, health checks, pods replication, and JWT middleware.
- Developed circuit breaker for state resilience. Implemented Kafka (publish-subscribe) for email verification.

EDUCATION

Carnegie Mellon University

Pittsburgh, PA, US

M.S. in Information Systems Management (Information Technology)

Sep 2021 – May 2023

Beijing University of Posts and Telecommunication

Beijing, China

Dual Degree: B.S. in Computer Engineering & Electronic Commerce

Sep 2017 – Jun 2021

CERTIFICATIONS & AWARDS

Tech Stack: Python, Java, Django, Linux, Databases, Netmiko, Nornir, TextFSM, Pyshark, Crontab/Celery, Redis

CCNA & CCNP: Cisco Certified Network Associate, Cisco Certified Network Professional

Patent: System and Apparatus for Natural Disaster Public Sentiment Analysis, CN No. 4648515, Granted in 2022

Nomination: The 38th CCF China Database Academic Conference (NDBC 2021)